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1. Selection sort

① {18, 22, 20, 19, 23}

→ { 18, 22, 20, 19, 21 }
Unsorted

$$\min = 18$$

swap the element 18 to its
right position

218, 22, 20, 19, 213
 sorted unsort

$$\min = 19$$

{ 18, 19, 20, 22, 21 }

sorted unsorted

$$\min = 20$$

{ 18, 19, 20, 22, 21, 3 }

sorted unsorted

→ 18, 19,

$$m/n = 2)$$

$\{ 18, 19, 20, 21, 22 \}$
 sorted unsorted

find ans

 $\{18, 19, 20, 21, 22\}$

$i = 0; j = 0; j < n - i - 1; j++ \}$
 $\text{int } j = 0; j < n - i - 1; j++ \}$

$> arr[j+1] \}$

② $\{34, 12, 56, 45, 23\}$

→ $\{34, 12, 56, 45, 23\}$

$\min = 12$

$\{ \underbrace{12}_{\text{sorted}}, \underbrace{34, 56, 45, 23}_{\text{unsorted}} \}$

$\min = 23$

$\{ \underbrace{12, 23}_{\text{sorted}}, \underbrace{56, 45, 34}_{\text{unsorted}} \}$

$\min = 34$

$\{ \underbrace{12, 23, 34}_{\text{sorted}}, \underbrace{45, 56}_{\text{unsorted}} \}$

$\min = 45$

$\{ \underbrace{12, 23, 34, 45}_{\text{sorted}}, \underbrace{56}_{\text{unsorted}} \}$

→ final ans

$\{12, 23, 34, 45, 56\}$

Time complexity $O(n^2)$

2. Bubble Sort

① { 30, 25, 27, 35, 29 }

Step 1 $\downarrow \leftrightarrow \downarrow$
{ 30, 25, 27, 35, 29 }

$30 > 25$

swap

{ 25, 30, 27, 35, 29 }

$30 > 27$, swap

{ 25, 27, 30, 35, 29 }

$30 < 35$, no swap

Step 2 $\downarrow \leftrightarrow \downarrow$
{ 25, 27, 30, 35, 29 }

$25 < 27$, no swap

Step 3 $\downarrow \leftrightarrow \downarrow$
{ 25, 27, 30, 35, 29 }

$27 < 30$ no swap

Step 4 $\downarrow \leftrightarrow \downarrow$
{ 25, 27, 30, 35, 29 }

$30 < 35$
no swap

Step 5 $\downarrow \leftrightarrow \downarrow$
{ 25, 27, 30, 35, 29 }

$29 < 35$ swap

int j=0; j<n-1
> arr[j+1] ?

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Step 6 $\downarrow \rightarrow \downarrow$
 $\{ 25, 27, 30, 29, 35 \}$

$30 > 29$ swap

Step 7 = $\{ 25, 27, 29, 30, 35 \}$

final ans

② $\{ 15, 10, 25, 40, 30 \}$

$\downarrow \rightarrow \downarrow$
 $\rightarrow \{ 15, 10, 25, 40, 30 \}$

$15 > 10$, swap.

$\{ 10, 15, 25, 40, 30 \}$

Step 2 $\downarrow \rightarrow \downarrow$
 $\{ 10, 15, 25, 40, 30 \}$

$15 < 25$ no swap.

Step 3 $\downarrow \rightarrow \downarrow$
 $\{ 10, 15, 25, 40, 30 \}$

$25 < 40$

no swap.

Step 4 $\downarrow \rightarrow \downarrow$
 $\{ 10, 15, 25, 40, 30 \}$
 $40 > 30$
swap

final ans

$\{ 10, 15, 25, 30, 40 \}$

Insertion Sort

① { 18, 22, 20, 19, 21 }

→ { 18, 22, 20, 19, 21 }

left < right = no shift

{ 18, 22, 20, 19, 21 }

left > right

shift

{ 18, 22, 22, 19, 21 }

{ 18, 20, 22, 19, 21 }

left > right

shift

{ 18, 20, 22, 22, 21 }

{ 18, 20, 19, 22, 21 }

left > right

shift

{ 18, 20, 19, 22, 21 }

{ 18, 20, 19, 21, 22 }

find ans

{ 18, 20, 19, 21, 22 }

② { 34, 12, 56, 45, 23? }

→ { 34, 12, 56, 45, 23? }

left > right

↓

shift

{ 34, 34, 56, 45, 23? }

{ 12, 34, 56, 45, 23? }

④ ~~right > left~~ left > right

↓

shift

{ 12, 34, 56, 56, 23? }

{ 12, 34, 45, 56, 23? }

left > right

↓

shift

{ 12, 34, 45, 56, 56? }

~~{ 12, 34, 45, 23 }~~

left > right

↓

shift

{ 12, 34, 45, 45, 56? }

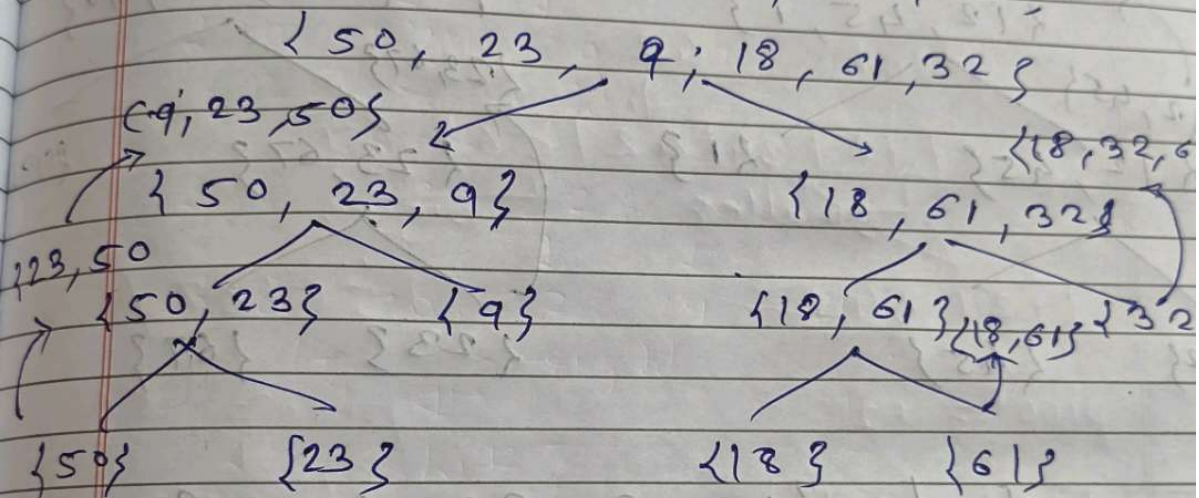
left > right

{ 12, 34, 34, 45, 56? }

{ 12, 23, 34, 45, 56 }

② Merge Sort

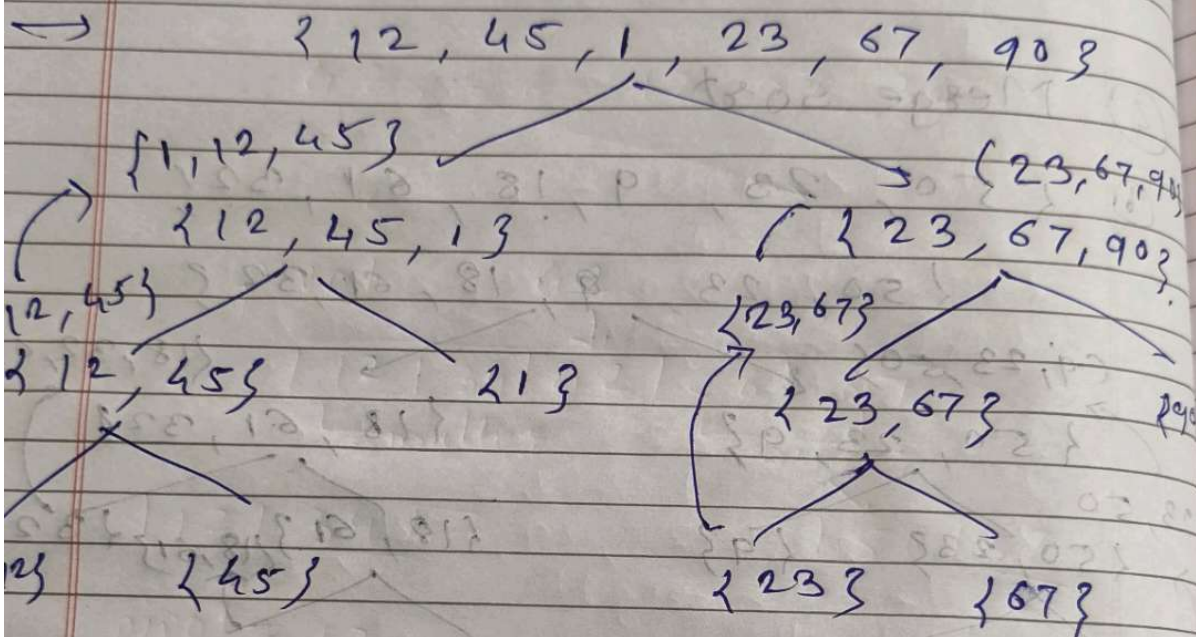
① { 50, 23, 9, 18, 61, 32 }



final = { 9, 18, 23, 32, 50, 61 }

$i = 0; j = n - i - 1; j++$?
 $(int j = 0; j < n - i - 1; j++)$?
 $arr[i] > arr[j+1]$?

② { 12, 45, 1, 23, 67, 90 }



final = { 1, 12, 23, 45, 67, 90 }

Q5

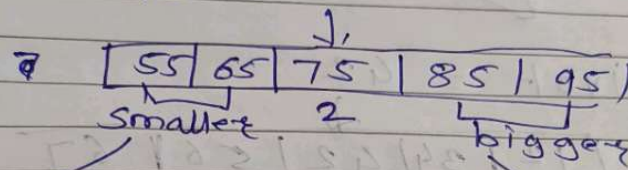
Quick sort-

①

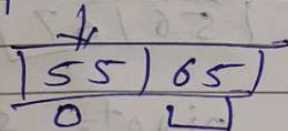
{ 75, 85, 55, 95, 65 }

→ 75, 85, 55, 95, 65

pivot = 75

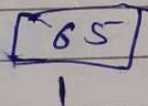


pivot = 55

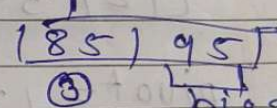


bigger

↓
pivot = 65

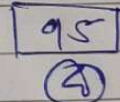


pivot = 85

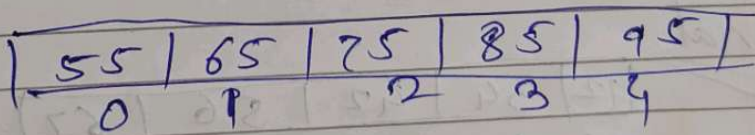


bigger

↓
pivot 95



final



② [42, 56, 12, 67, 34]

→

~~42, 56, 12, 67, 34~~

pivot = 42

↓
[12 | 34 | 42 | 56 | 67]
smaller bigger

[12 | 34]

[56 | 67]

pivot = 12

↓
[12 | 34]
bigger
↓

pivot = 34

↓
[34]

pivot = 56

↓
[56 | 67]
bigger

pivot = 67

↓
[67]

final

[12 | 34 | 42 | 56 | 67]
0 1 2 3 4